

Matrix Theory: Lecture 1

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Outline

- General course information
- Inverses
- Some special matrix classes
- LU-factorization & Gauss Elimination
- Linear independence and span.



Something about me

Carl Olsson

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- Dissertation 2009.
- Mathematics LTH, the Computer Vision and Machine Learning division.
- Research within computer vision.



General Course Information

Five versions of the course:

- FMAN70, 6 hp, (Grades: U,3,4,5).
- FMA121F, 6 hp (for PhD students, Grades U,G).
- FMAN71 , 7.5 hp (Grades: U,3,4,5).
- MATC70 (Faculty of science), 7.5 hp (Grades U,G,VG).
- FMAN71F, 7.5 hp (for PhD students, Grades U,G).

Content

- FMAN71, FMAN71F and MATC70 all the chapters Holst-Ufnarovski.
- FMAN70 and FMA121F excludes chapter 10.



General Course Information

Course requirements

- Two mandatory computer assignments (completed before exam).
- Written home exam followed by oral exam.

Canvas Page

<https://canvas.education.lu.se/courses/36924>

Timeedit

<https://cloud.timeedit.net/lu/web/lth1/ri1X80g61560Y2QQ75Z68X3Y0Yy7003355760Q568f525Z4377QY.html>



Sign-up and Registration

If you have not signed up for the course or have registration issues contact the administrators: expedition@math.lth.se



CEQ-report and Student Representatives

FMAN71_2024_HT_LP2_arbetsrapport_en.pdf

Interested in being a student representative for the course?

Contact me: carl.olsson@math.lth.se



A theoretical course

Expectations:

- The course focuses on theory and proofs. (First course to do so?)
This requires a lot of work if you are not used to it.
- It will not be enough to learn algorithms, one needs to understand the theory and explain why an approach works.
- Real problems are often too large to solve by hand (we leave it to the computers). We just need to understand the theory and prove that an algorithm/method will work for a given problem.

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Feedback Opportunities

- Voluntary exercise sessions. Need to ask questions to get feedback!
- Voluntary home assignments. Feedback on your written solutions.
- 2 mandatory computer assignments

Note: The program contains a lot of exercises. These are voluntary, and you do not have to do all of them. **Select the ones that you think you will benefit from!**



Outline

- Inverses
- Some special matrix classes
- LU-factorization & Gauss Elimination
- Linear independence and span.



Inverses

Def. A matrix L is a left inverse to A if $LA = I$.

A matrix R is a right inverse to A if $AR = I$.

A^{-1} is a matrix that is both left and right inverse to A .

Theorem 1.8

If A, B are invertible then $(AB)^{-1} = B^{-1}A^{-1}$. Analogous results hold for left and right inverses.



Nilpotent Matrices

A ($n \times n$) is nilpotent if $A^k = AA \cdots A$ (k times) $= 0$.

Ex. (Upper and Lower) triangular matrices with zero diagonal.

$$\begin{pmatrix} 0 & 0 & 0 & 0 \\ * & 0 & 0 & 0 \\ * & * & 0 & 0 \\ * & * & * & 0 \end{pmatrix} \begin{pmatrix} 0 & 0 & 0 & 0 \\ * & 0 & 0 & 0 \\ * & * & 0 & 0 \\ * & * & * & 0 \end{pmatrix} = \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ * & 0 & 0 & 0 \\ * & * & 0 & 0 \end{pmatrix}.$$
$$\begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ * & 0 & 0 & 0 \\ * & * & 0 & 0 \end{pmatrix} \begin{pmatrix} 0 & 0 & 0 & 0 \\ * & 0 & 0 & 0 \\ * & * & 0 & 0 \\ * & * & * & 0 \end{pmatrix} = \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ * & 0 & 0 & 0 \end{pmatrix}.$$

Nilpotent with $A^4 = 0$.

Theorem 1.7

If A is nilpotent then $I - A$ is invertible and $(I - A)^{-1} = I + B$ where B is nilpotent.

Block Matrix Multiplication

If the sizes of the blocks are right we can multiply the blocks.
Same principle as regular matrix multiplication:

$$\begin{pmatrix} X & Y \\ Z & T \end{pmatrix} \begin{pmatrix} U & V \\ W & R \end{pmatrix} = \begin{pmatrix} XU + YW & XV + YR \\ ZU + TW & ZV + TR \end{pmatrix}$$

X and Z has the same number of columns as U and V has rows,
 Y and T has the same number of columns as W and R has rows.



Triangular Matrices

Theorem 1.9

If T is triangular with non-zero diagonal elements then

- 1 T^{-1} exists and is triangular.
- 2 $\begin{pmatrix} T \\ B \end{pmatrix}$ has left inverse $(T^{-1} \ 0)$ for any B
- 3 $\begin{pmatrix} T & C \end{pmatrix}$ has right inverse $\begin{pmatrix} T^{-1} \\ 0 \end{pmatrix}$ for any C .



Gaussian Elimination

From Linear Algebra:

Solve $AX = B$ by:

- 1 Find upper triangular (echelon/staircase form) matrix U such that

$$AX = B \Leftrightarrow UX = C$$

- 2 Solve $UX = C$ from last equation to first.

Formal definition

First element of a non-zero row is called pivot element.

A matrix U is in **echelon form** if it fulfills:

- 1 if row i , with $i > 1$, is non-zero, then the row above is also non-zero.
- 2 if u_{ij} , u_{i-1k} are pivot elements then $k < j$.

The first r rows of U are non-zero, the remaining ones are zero.

The **rank** of U is r .



Gaussian Elimination

Ex.

$$U = \begin{pmatrix} 0 & \underline{2} & 4 & 6 & 8 \\ 0 & 0 & \underline{1} & 2 & 3 \\ 0 & 0 & 0 & 0 & \underline{1} \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix} \quad C = \begin{pmatrix} 0 \\ 2 \\ 1 \\ c_4 \end{pmatrix}$$

The system has solution if and only if $c_4 = 0$.

Solve for x_5 from row 3

Set $x_4 = t$, solve for x_3 from row 2.

Solve for x_2 from row 1.

Set $x_1 = s$.

Underlined elements are **pivot elements**

Columns with pivot elements are **pivot columns**

x_2, x_3, x_5 are **basic variables** (since they correspond to pivot columns)

x_1, x_4 are **free variables**



Gaussian Elimination and LU-factorization

In Gaussian Elimination we perform row operations (and possible row permutations).

Can be written as matrix multiplications with permutation- and elementary- ($I + cE_{ij}$) matrices.

Theorem 2.1 (LU-factorization)

For every matrix A there exist a permutation matrix P , an echelon matrix U and a lower triangular matrix L with ones on the main diagonal such that

$$PA = LU.$$



Linear Independence

Def. $A_1, A_2, \dots, A_n$ is linearly independent if

$$x_1 A_1 + x_2 A_2 + \dots + x_n A_n = 0 \Rightarrow x_1 = x_2 = \dots = x_n = 0$$

In matrix form: If $A = (A_1 \ A_2 \ \dots \ A_n)$

$$AX = 0 \Rightarrow X = 0$$

Theorem 2.6

For $A - (m \times n)$ the following is equivalent:

- ① A has left inverse
- ② $AX = B$ has at most one solution
- ③ The columns of A are linearly independent
- ④ $r(A) = n$

Span

Def. $A_1, A_2, \dots, A_n$ span/generate $\mathbb{K}^m$ if every vector $B \in \mathbb{K}^m$

$$B = x_1 A_1 + x_2 A_2 + \dots + x_n A_n.$$

In matrix form: If $A = (A_1 \ A_2 \ \dots \ A_n)$

$$AX = B$$

has solution for every B .

Theorem 2.8

For $A - (m \times n)$ the following is equivalent:

- ① A has right inverse
- ② $AX = B$ has at least one solution for each B
- ③ The columns of A span $\mathbb{K}^m$
- ④ $r(A) = m$

Inverses and square matrices

Corollary

If A has an inverse A^{-1} then A is square.

Proof. Suppose A is $m \times n$:

A has left inverse $A^{-1} \Rightarrow r(A) = m$.

A has right inverse $A^{-1} \Rightarrow r(A) = n$.

Therefore $m = n$.

